

Project- Evizon

(From Chaos to Clarity)

1. Problem Statement

Universities conduct numerous events—seminars, cultural programs, sports meets, workshops, fests—generating thousands of photographs every semester. These images are essential for institutional documentation, NAAC reports, newsletters, websites, social media, and archival history.

However, event photos often remain **unorganized, unlabeled, and stored randomly** across devices, drives, and folders.

This leads to:

- Loss of important documentation
- Manual sorting taking hours or days
- Difficulty identifying which photo belongs to which event
- Repetitive, tedious human effort
- Poor archival quality and information retrieval challenges

There is a clear need for an **AI-driven automated system** that can tag, group, categorize, summarize, and clean event photos efficiently.

2. Relevance to Market / Society / Industry

The education sector is rapidly digitalizing. Institutions increasingly rely on multimedia assets for:

- NAAC/NBA accreditation
- University newsletters
- Social media outreach
- Annual reports
- Websites and magazines
- Alumni documentation

Manual image organization is inconsistent and unsustainable.
An automated solution like EVizon addresses this gap by enabling:

- Fast documentation workflows
- Zero manual tagging effort
- High-quality archiving
- Easy photo retrieval
- Professional digital asset management

Such a system is valuable for **schools, colleges, universities, event companies, brands, and media teams** worldwide.

It aligns with the growing demand for **AI-based multimedia processing solutions**.

3. Proposed AI-Based Solution

Evizon is an AI-powered Event Image Tagging & Archival System that automates the entire photo management pipeline.

Users upload multiple event photos - Evizon processes them automatically using Vision AI.

The system performs:

Event Type Detection

Identifies whether the image belongs to:

- Seminar / Workshop
- Cultural Event
- Sports Event
- Orientation / Farewell
- Award Ceremony
- Classroom Activity

Automatic Tag Generation

Detects:

- Objects: stage, mic, banner, projector, trophies, sports items
- Activities: speaking, dancing, presenting, playing
- Scene type: indoor/outdoor, auditorium/ground/classroom

Face Clustering (Not Recognition)

Groups photos containing the **same faces**, without identifying individuals. Useful for selecting photos featuring speakers, performers, awardees, etc.

Duplicate & Blurry Image Detection

- Removes duplicate photos
- Flags low-quality/blurry images
- Improves final archive quality

AI-generated Event Summary

Automatically generates a short description of the event using detected patterns.

Event-wise Album Creation

Creates structured folders/albums for:

- Event name
- Location
- Date (if metadata available)
- People clusters

Evizon removes the need for manual photo sorting and accelerates university-level documentation.

4. Objectives

1. To design a Vision-AI tool that automatically identifies and categorizes university event photographs.
2. To implement multi-label tagging for objects, activities, and scenes.
3. To cluster similar faces for simplified image grouping.
4. To detect and filter duplicate or blurry images for improved quality.
5. To auto-generate concise event summaries based on image-level insights.
6. To provide a simple user interface for uploading, viewing, and downloading organized albums.
7. To create a scalable digital archive system for educational institutions.

5. Implementation Plan / Methodology

Phase 1 – Dataset & Preprocessing

- Use Google Open Images Dataset + sample university photos
- Extract object and scene labels
- Preprocess for uniform resolution

Phase 2 – AI Model Integration

- Use pre-trained models from HuggingFace for:
 - Event classification
 - Object/activity detection
 - Scene recognition
- Use MediaPipe for face embeddings

Phase 3 – Clustering & Quality Filtering

- DBSCAN/KMeans for face clustering
- Laplacian variance for blur detection
- Perceptual hashing for duplicate detection

Phase 4 – Album Generation

- Combine all detections and group photos by:
 - Event type
 - Scene
 - Face clusters
 - Quality score

Phase 5 – Summary Generator

- Feed detected tags into a small LLM for:
 - 2–3 line event summary

Phase 6 – UI & Dashboard

- Simple web UI: upload → process → organized albums
- Downloadable folders
- Event summary display

Phase 7 – Testing & Refinement

- Test with real campus photos
- Optimize tagging accuracy
- Improve clustering precision

6. Innovation & Uniqueness

Evizon stands out due to the combination of **five high-impact AI capabilities** rarely integrated together:

1. **Automated Event Classification**
2. **Advanced Multi-label Tagging**
3. **Face Clustering (Privacy-Safe)**
4. **Automatic Quality Control** (blur + duplicate removal)
5. **AI-based Event Summary Generation**

Unlike basic photo organizers, Evizon provides a **complete end-to-end intelligent archival system**, specifically designed for university documentation workflows.

7. Expected Outcomes

- A functional prototype capable of reading, tagging, and grouping event images.
- Higher accuracy in event categorization and object detection.
- Organized digital albums ready for university-level archival use.
- Reduction in manual photo sorting effort by 80–90%.
- A searchable event archive with AI-generated descriptions.

8. Expected Impact

Institutional Impact

- Improved documentation quality
- Faster content creation
- Streamlined NAAC/NBA reporting
- Better preservation of campus history

Social Impact

- Availability of organized event media for students and alumni
- Transparency and accessibility in digital documentation

Technology Impact

- Promotes adoption of Vision-AI
- Demonstrates real-world application of machine learning in education

9. Future Scope

- Integration with cloud storage (Google Drive / AWS S3)
- Multi-event timeline visualization
- Automatic video clipping + scene detection
- Integration with college CMS systems
- Support for event-level face recognition (optional, privacy-controlled)
- Cross-campus deployment as a full SaaS product